

WASP-10b

R-1295

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_191002 · created 2026-07-31T16:40:15+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458758.8117 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1464 +/- 0.0089
Transit depth (Rp/Rs)^2	2.14 +/- 0.26 [%]
Orbital Inclination (inc)	89.9 +/- 0.61
Airmass coefficient 1 (a1)	0.9642 +/- 0.0097
Airmass coefficient 2 (a2)	0.031 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.01 %
Best Comparison Star	#2 - [347, 139]
Optimal Aperture	5.0
Optimal Annulus	14.54
Transit Duration (day)	0.0936 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1464 ± 0.0089 vs 0.1592 (deviation 1.44σ)
Duration fit vs published	0.0936 d vs 0.0946 d (deviation 0.89σ)
Mid-time O–C	2.5 min
Depth SNR	16.4
Residual scatter	1.01 %

Light curve

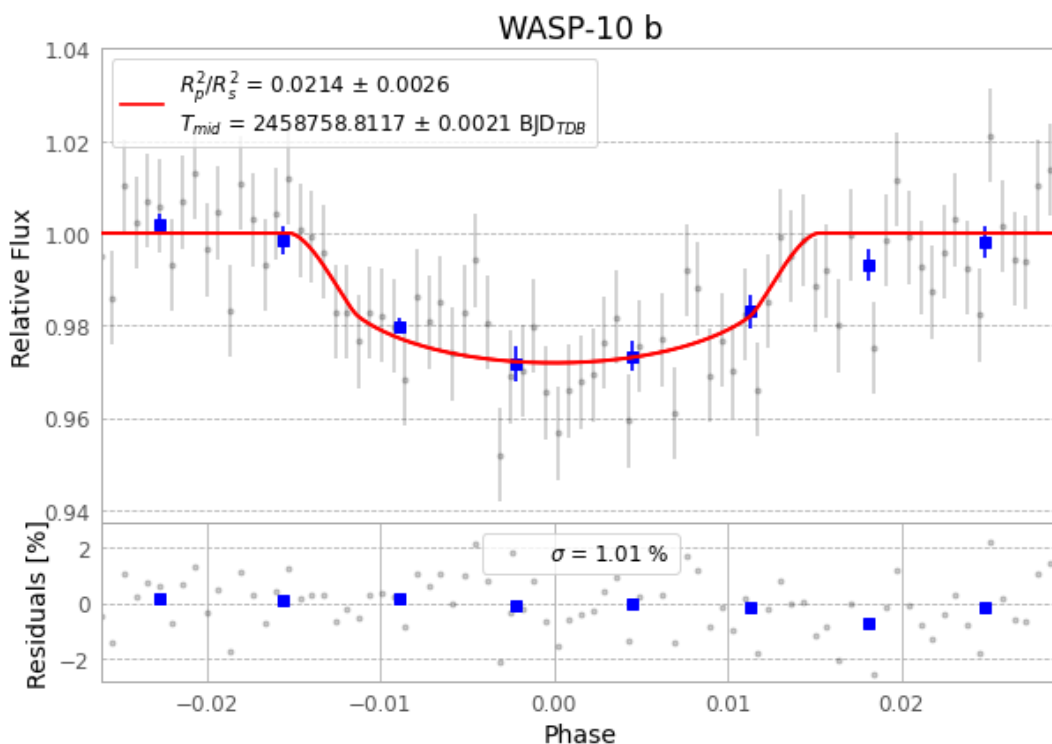


Figure 1 — FinalLightCurve_WASP-10 b_2019-10-01.png (SHA-256 21fe37bc1bb853b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [350, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7df030705a904ec4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-10191002052412.FITS → WASP-10191002093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-191002.log (bb49f2c9e97e...), FinalLightCurve_WASP-10 b_2019-10-01.png (21fe37bc1bb8...), FinalLightCurve_WASP-10 b_2019-10-01.csv (676f44be8db8...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 16:40Z.